Code: 23A54101 **R23**

B.Tech I Year I Semester (R23) Supplementary Examinations June/July 2024

LINEAR ALGEBRA & CALCULUS

(Common to all branches)

Time: 3 hours Max. Marks: 70

PART – A

(Compulsory Question)

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1 Answer the following: (10 X 02 = 20 Marks)

(a) Find the value of k such that the rank of $\begin{bmatrix}1 & 2 & 3\\ 2 & k & 7\\ 3 & 6 & 10\end{bmatrix}$ is 2 --- 2M

(b) If the Echelon form of the matrix A = $\begin{bmatrix}1 & 2 & 3\\ 0 & 3 & 7\\ 0 & 0 & c-3\end{bmatrix}$ and if C = 3. Then the rank of the matrix is equal to ______________ 2M

(c) The Eigen values of $\begin{bmatrix}2 & \sqrt{2}\\ \sqrt{2} & 2\end{bmatrix}$. 2M

(d) Obtain a real symmetric matrix of the quadratic form $3x^2 - 2y^2 - z^2 - 4xy + 12yz + 8xz$. 2M

(e) Verify Whether Rolles theorem can be applied to the function f(x) = x^3 in[1, 3]. 2M

(f) Obtain Taylor's expansion of e^x about x = -1. 2M

(g) Find first and second order partial derivatives of $x^3 + y^3 - 3axy$. 2M

(h) If u = 3x + y , v = x - 2y, then $\frac{\partial(u,v)}{\partial(x,y)} =$ 2M

(i) Transform the given integral into polar coordinates 2M

$$\int_0^\infty \int_0^\infty e^{-(x^2+y^2)}\ dx\, dy$$

(j) Change the order of Integration for the integral $\int_0^2 \int_1^{e^x} dy\, dx$. 2M

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

2 Find the Inverse of A = $\begin{bmatrix}-2 & 1 & 3\\ 0 & -1 & 1\\ 1 & 2 & 0\end{bmatrix}$ using Gauss Jordan method. 10M

OR

3 Check for the consistency. If so solve the system of linear equations . 10M
x + 2y + z = 3, 2x + 3y + 2z = 5, 3x - 5y + 5z = 2 and 3x + 9y – z = 4.

4 Verify Cayley-Hamilton theorem and hence find A^{-1} and A^4. 10M
if A = $\begin{bmatrix}7 & 2 & -2\\ -6 & -1 & 2\\ 6 & 2 & -1\end{bmatrix}$.

OR

5 Reduce the Quadratic form to the canonical form $3x^2 - 3y^2 - 5z^2 - 2xy - 6yz - 6xz$. 10M

Contd. in Page 2

6	(a)	Verify Rolle's theorem for f(x) = $x^3 - 2x - 3$ in the interval (1, -3).	5M
	(b)	Prove that $\left(x - \frac{x^2}{2}\right) < \log(1+x) < \frac{x^2}{2(1+x)}$ for x > 0.	5M
		OR	
7	(a)	Verify Cauchy's Mean value theorem for the functions f(x) and $f^1(x)$ in(1, e) given f(x)=log x.	5M
	(b)	Find the Taylors series expansion of sin2x about x = π/4.	5M
8	(a)	Use lagrange's method of undetermined multipliers to find the maximum or minimum of $x^2 + y^2 + z^2$ given that z^2 = 1 + xy.	5M
	(b)	$f(x) = x + 3y^2 - z^3;\ g(x,y,z) = 2x^2yz;\ h(x,y,z) = 2z^2 - xy\ , then\ find\ \frac{\partial(f,g,h)}{\partial(x,y,z)}$.	5M
		OR	
9	(a)	Find the dimensions of rectangular box of maximum capacity whose surface area is S.	5M
	(b)	Find $\frac{du}{dx}$ if u = sin($x^2 + y^2$) , where $a^2x^2 + b^2y^2 = c^2$.	5M
10		Find the Volume bounded by the plane, the cylinder $x^2 + y^2 = 1$ and the plane x + y + z = 3.	10M
		OR	
11		Evaluate $\iint (x^2 + y^2)\, dx\, dy$ over the area bounded by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.	10M

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